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Planar vessel mixing using a brushless drive surrounding the reaction space

Abstract

A planar vessel mixing system is disclosed. The mixing system may include a mixing vessel, which may be a filter reactor, with a mixing region for receiving mixing substances, which may be liquid or solid. The vessel may include at least one magnetic stir bar to mix the substances. At least one brushless magnetic drive may be disposed externally around the mixing vessel and may be configured to generate a rotating magnetic field to rotate the magnetic stir bar, thereby mixing the mixing substances. The brushless magnetic drive may be adjustable along a length of the mixing vessel. A filter may be disposed at the bottom of the mixing vessel to filter out byproducts from the mixing. Use of the brushless magnetic drive may enable small-scale mixing (e.g., less than 100 mL) by removing the need for an overhead stirrer.

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Background/Summary

BACKGROUND

1. Field

(1) Embodiments of the current disclosure generally relate to mixing in small-scale vessels (e.g., having a volume of less than 100 mL). More specifically, embodiments of the current disclosure

relate to non-contact planar mixing using a brushless magnetic drive that generates a rotating magnetic field to drive a magnetic stir bar in a vessel to induce mixing.

2. Related Art

(2) Small-scale vessels, such as tabletop filter reactors with bottom drain ports and mixing volumes less than 100 milliliters, are often used for continuous flow chemistry. Specifically, vessels of this scale are frequently used for liquid-liquid and solid-liquid multiphase systems, automated experimentation, chemical exploration, and expensive materials, where using larger vessels is overly wasteful. Currently, mixing of heterogeneous systems in small-scale vessels is typically performed either in standard lab glassware, followed by material transfer and manual filtration, or in a dedicated filter-bottom reactor. It is often desirable for such systems to filter solids (e.g., crystals or other solids) in a dedicated filter-bottom reactor, while filtering out the remaining liquids, often via a bottom-mounted drain port. The presence of a bottom drain port within such glassware requires overhead agitation where an agitator (e.g., a stirrer) is inserted into the mixing volume from the top of the mixing vessel. However, the use of overhead stirring imposes additional spatial limitations on the vessels in order to accommodate the size of the overhead stirring apparatus. Mixing vessels with overhead stirring units and bottom filters are typically limited to a minimum working volume of 150 milliliters. As discussed above, the use of larger vessels can lead to excessive waste of materials, which can be prohibitive in applications such as pharmaceuticals, where reactants include expensive or scarce substances. Additionally, leakage may occur where the overhead stirrer apparatus is connected to the vessel. In some instances, the substances being mixed are dangerous, such that a leak could be dangerous.

(3) Some small-scale vessels utilize a tabletop magnetic stir plate that stirs a magnetic stir bar located in the mixing vessel. However, the presence of the aforementioned bottom drain ports or filtration systems prevents the use of magnetic stir plates. Further, such systems typically rely on the manual intervention in the process by an operator (for example, to manually filter the products of the reaction) and are generally limited to reactions conducted at atmospheric pressure, since the vessel must be opened at multiple points in the chemical process for operator access. Such actions are of particular concern for air-sensitive experiments, as opening a vessel multiple times for manual intervention may increase the risk of harmful exposure the reactants and the reaction environment to air. As such, what is needed is a system for small-scale mixing (e.g., less than 100 mL) that obviates the need for an overhead stirrer.

SUMMARY

(4) Embodiments of the current disclosure solve the above-described problems and provide a distinct advancement in the art by providing systems and methods for non-contact mixing for small-scale filter reactors, such as those having a volume of less than 100 mL. Planar vessel mixing systems, where the stirring action is in a single plane, may comprise a mixing vessel with a magnetic stir bar, such as a filter reactor, and a non-contact, brushless magnetic drive for driving the magnetic stir bar. The brushless magnetic drive may be positioned external to the mixing vessel and may generate a rotating magnetic field that rotates the magnetic stir bar.

(5) The systems described herein may enable mixing in small-scale filter reactors, which may enable continuous processes via integration into flow chemistry at a tabletop scale. Moreover, the contactless mixing systems described herein may be integrated into pressurized systems, allowing for a wider range of operating temperatures and pressures at which processes can occur. Magnetic stirring of the mixing substances eliminates the need for overhead agitation that burdens systems with additional pressurization and spatial requirements. As a result, filter reactor systems and other vessels may be configured to avoid batch synthesis and to increase the throughput and overall yield of the process. Moreover, the capacity of the mixing vessel for pressurization permits operators to run reactions at higher temperatures, allowing for a broader range of potential solvents to be used, thus expanding the type of reactions and subsequent filtration that the mixing system can execute.

(6) In some aspects, the techniques described herein relate to a mixing system, including: a mixing

vessel, including: a fluid inlet; a mixing region configured to receive a first mixing substance and a second mixing substance via the fluid inlet; and at least one outlet fluidly connected to the mixing region; a magnetic stir bar disposed within the mixing vessel; a brushless magnetic drive that surrounds the mixing vessel and that is configured to generate a rotating magnetic field, thereby rotating the magnetic stir bar to mix the first mixing substance with the second mixing substance; and a controller configured to control a speed and a direction of the rotating magnetic field.

(7) In some aspects, the techniques described herein relate to a system for vessel mixing, including: a pressurizable mixing vessel, including: at least one inlet; a mixing region configured to receive a first mixing substance and a second mixing substance via the at least one inlet; and at least one outlet fluidly connected to the mixing region; a magnetic stir bar disposed within the pressurizable mixing vessel; and a brushless magnetic drive surrounding the pressurizable mixing vessel and configured to generate a rotating magnetic field, thereby rotating the magnetic stir bar to mix the first mixing substance with the second mixing substance.

(8) In some aspects, the techniques described herein relate to a system for planar vessel mixing, including: a mixing vessel, including: a fluid inlet; a mixing region configured to receive a first mixing substance and a second mixing substance via the fluid inlet; and at least one outlet fluidly connected to the mixing region; a magnetic stir bar disposed within the mixing vessel; a first brushless magnetic drive surrounding the mixing vessel and operable to generate a first magnetic field that rotates a first magnetic stir bar to thereby mix the first mixing substance and the second mixing substance; and a second brushless magnetic drive that surrounds the mixing vessel.

(9) This summary is provided to introduce a selection of concepts in a simplified form that are further described below in the detailed description. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to limit the scope of the claimed subject matter. Other aspects and advantages of the present disclosure will be apparent from the following detailed description of the embodiments and the accompanying drawing figures.

Description

BRIEF DESCRIPTION OF THE DRAWING FIGURES

(1) Embodiments of the present disclosure are described in detail below with reference to the attached drawing figures, wherein:

(2) FIG. 1A depicts a schematic view of a planar vessel mixing system for some embodiments;

(3) FIG. 1B depicts an isometric view of the planar vessel mixing system for some embodiments;

(4) FIG. 1C depicts a top perspective view of the planar vessel mixing system for some embodiments;

(5) FIG. 2 depicts a planar vessel mixing system with multiple brushless magnetic drives for some embodiments; and

(6) FIG. 3 depicts a planar vessel mixing system with a double-walled heating element.

(7) The drawing figures do not limit the present disclosure to the specific embodiments disclosed and described herein. The drawings are not necessarily to scale; emphasis is instead placed upon clearly illustrating the principles of the present disclosure.

DETAILED DESCRIPTION

(8) The following description of embodiments of the present disclosure references the accompanying illustrations that illustrate specific embodiments in which the present disclosure can be practiced. The embodiments are intended to describe aspects of the invention in sufficient detail to enable those skilled in the art to practice the present disclosure. Other embodiments can be utilized, and changes can be made without departing from the scope of the present disclosure. The following detailed description is, therefore, not to be taken in a limiting sense.

(9) In this description, references to “one embodiment,” “an embodiment,” “embodiments,”

“various embodiments,” “certain embodiments,” “some embodiments,” or “other embodiments” mean that the feature or features being referred to are included in at least one embodiment of the technology. Separate references to “one embodiment,” “an embodiment,” “embodiments,” “various embodiments,” “certain embodiments,” “some embodiments,” or “other embodiments” in this description do not necessarily refer to the same embodiment and are also not mutually exclusive unless so stated and/or except as will be readily apparent to those skilled in the art from the description. For example, a feature, structure, act, etc., described in one embodiment may also be included in other embodiments but is not necessarily included. Thus, the current technology can include a variety of combinations and/or integrations of the embodiments described herein.

(10) Generally, embodiments of the current disclosure are directed to a planar vessel mixing system for control of continuous chemical processes of small-volume solutions, such as liquid-liquid mixing and liquid-solid mixing. As used herein, a small volume or small vessel refers to any volume or vessel having a mixing volume of less than 100 milliliters. It will be appreciated that embodiments of the present disclosure are not limited to mixing volumes less than 100 milliliters, and that, generally, any mixing volume may be used. As discussed further herein, the disclosed embodiments may be advantageous for use with filter reactors that are typically limited to a minimum working volume of 150 milliliters, as these filter reactors must fit an overhead stirring unit, along with loading ports for supplying the mixing constituents into the mixing volume.

(11) More specifically, some embodiments of the current disclosure are directed to magnetic agitation of filtration vessels, including filter reactors that require agitation but experience spatial limitations for an agitator or mixer due to the presence and location of filters. The planar vessel mixing system may include a brushless magnetic drive that may be positioned externally to the mixing vessel and may generate a rotating magnetic field to spin a magnetic mixer (e.g., a magnetic stir bar) located within the mixing vessel for non-contact mixing. The brushless magnetic drive may include a stator that generates an alternating magnetic field, with the magnetic stir bar functioning as a rotor. The brushless drive may surround the vessel at various vertical positions of the mixing region, allowing for thorough agitation of mixing substances at various locations within the mixing region. Thus, the drive system may encompass the reaction space of the filter reactor in contrast to prior drive systems that are overhead the filter reactor, leading to the above-mentioned spatial and continuous flow chemistry problems by providing a planar vessel mixing system that enables integration of tabletop scale flow chemistry in chemical processes, which may be useful in continuous manufacturing systems that utilize mixing volumes less than 100 milliliters.

(12) As discussed, the mixing vessel may be a filter reactor, which is a mixing vessel in which a chemical reaction can occur through the mixing of multiple substances or reactants, and that includes a filter for filtration of the resulting reaction products. The mixing substances enter the filter reactor via inlet ports at the top of the filter reactor. Following the reaction, any solid products of the reaction are filtered out to prevent them from passing through to the next component of the manufacturing system along with the desired solution.

(13) FIGS. 1A, 1B, and 1C illustrate a schematic view, an isometric view, and a top perspective view, respectively, of a mixing system **100** in accordance with embodiments of the present disclosure. Mixing system **100** may include a mixing vessel **102**, which may be a filter reactor. Mixing vessel **102** may be generically cylindrical or may take various other geometries (e.g., rectangular prism, triangular prism, trapezoidal prism, or having a conical or round body with a cylindrical neck) without departing from the scope hereof. In some embodiments, the mixing vessel **102** is configured for small-scale, tabletop chemical processes in continuous manufacturing systems. As used herein, the term “small-scale,” may mean that mixing vessel **102** has an internal volume of about 1 milliliter to about 100 milliliters, or of about 1 milliliter to about 50 milliliters, or of about 1 milliliter to about 25 milliliters. In some embodiments, the mixing vessel **102** may have an internal volume of about 10 milliliters to about 20 milliliters. In some embodiments, the mixing volume may be less than 10 milliliters, less than 100 milliliters, less than 1 liter, or less than

10 liters. It will be appreciated that mixing volumes of greater than 100 milliliters (e.g., about 500 milliliters, about 1 liter, about 5 liters, etc.) are within the scope hereof. As previously discussed, filter reactors are typically constrained to having volumes of at least 150 milliliters to accommodate an overhead stirring apparatus; thus, providing mixing vessel **102** capable of mixing substances in a mixing volume of less than 100 milliliters is advantageous to enable mixing on a smaller scale than is possible with overhead stirrers.

(14) Mixing vessel **102** may include a mixing region **104a** in which substance **106a** and substance **106b** may be mixed. Although two substances are described herein for brevity, any number of substances may be mixed within the scope of the present disclosure. The mixing vessel **102** may be configured to accommodate continuous processes involving both chemical reaction and solution filtration within the same vessel in a mixing region **104a**. The mixing region **104a** may be vertically bounded by a top surface **104b** of the mixing vessel **102** and a bottom surface **104c** of the mixing vessel **102**. The distance between top surface **104b** and bottom surface **104c** may define the length of the mixing vessel **102**.

(15) The mixing region **104a** may receive substance **106a** and substance **106b** via inlet ports **108**. Inlet ports **108** may be fluidly connected to the mixing region **104a**. In some embodiments, mixing vessel **102** comprises a combined inlet/outlet port through which substances **106a**, **106b** enter and exit. Each substance, such as substance **106a** and substance **106b**, may comprise a reactant or, more generally, a component (also referred to as mixing constituents) added to the mixing vessel **102** for a chemical reaction. For example, substance **106a** and substance **106b** may be pharmaceutical compounds that are to be mixed in mixing region **104a**. The inlet ports **108** may be coupled to the mixing vessel **102** proximate to the mixing region **104a** in some embodiments.

(16) The mixing vessel **102** may be configured to operate at non-atmospheric pressures and may, as such, be pressurizable (e.g., pressure-tight and having a sufficient tensile strength to contain a desired operating pressure). Accordingly, in some embodiments, at least one pump such as pump **110** may be provided to pressurize the mixing vessel **102**, allowing for the integration of the vessel into pressurized systems operating at non-atmospheric pressures. Other pressure sources may be employed. In some embodiments, the mixing vessel **102** may be pressure-rated up to 150 pounds per square inch. Pump **110** may also be configured to pump in one or more of substance **106a** and substance **106b** into the mixing region **104a**. In some embodiments, multiple instances of pump **110** are provided, such as one pump for each of substance **106a** and substance **106b** that is fed into the mixing vessel **102**.

(17) Additionally, in some embodiments, inert gases, such as nitrogen, may be added to the headspace of mixing vessel **102** (i.e., the region above substance **106a** and substance **106b** in mixing region **104a**) in order to prevent oxidation during continuous manufacturing system reactions. Thus, the chemical reaction and subsequent filtration may occur without exposing the mixture to open air, preventing oxidation and contamination during the continuous chemical process. In some such embodiments, the inlet ports **108** may be used to fill the headspace of the mixing vessel **102** with inert gases. For example, when mixing vessel **102** includes three instances of inlet ports **108** as shown, two of the instances of inlet ports **108** may be used to deliver substance **106a** and substance **106b** into mixing region **104a**, and the third of the instances of inlet ports **108** may be used to deliver the inert gases into the mixing vessel **102**.

(18) In some embodiments, the mixing vessel **102** may be constructed of glass, ideally structured to allow for continuous observation and for achieving a sufficient pressure rating for non-atmospheric pressures. However, the mixing vessel **102** may also be formed from plastics, metals, or alloys. In some embodiments, the mixing vessel **102** may be constructed of metals such as stainless steel (e.g., 316 stainless steel, 316L stainless steel, etc.). In some embodiments still, the mixing vessel **102** may be formed from stainless steel alloys (e.g., alloy A-286, 20, 230, 400, 600, 625, B-2/B-3, C-276, etc.) without departing from the scope hereof. Generally, mixing vessel **102** may be formed of any material.

(19) To mix substance **106a** with substance **106b**, the mixing system **100** may further include at least one instance of magnetic drive **112** that may be coupled to mixing vessel **102** and disposed around the exterior (e.g., the circumference) of the mixing vessel **102**. The magnetic drive **112** may, in some embodiments, surround the mixing vessel **102** such that the magnetic drive **112** is concentric with the mixing vessel **102** (see FIG. 1C). In some embodiments, the magnetic drive **112** may be powered by an external power source such as, but not limited to, a battery (not shown). Generally, magnetic drive **112** may be located at any vertical position along the exterior of the mixing vessel.

(20) The magnetic drive **112** may be energized in an alternating circular pattern such that a rotating magnetic field **114** is produced. The rotating magnetic field **114** may then rotate a magnetic stir bar **116** located within the mixing region **104a**. The magnetic stir bar **116** may be positioned in mixing region **104a** and may be suspended at a vertical position within mixing vessel **102** that corresponds with the vertical position of magnetic drive **112**. That is, the rotating magnetic field **114** may align or hold the magnetic stir bar **116** therewith. Thus, the combination of magnetic drive **112** and magnetic stir bar **116** stir bar forms a brushless DC motor, with magnetic drive **112** forming the stator, and magnetic stir bar **116** forming the rotor.

(21) In some embodiments, the magnetic stir bar **116** is generally cylindrical or stadium-shaped, as shown. However, it will be appreciated that the magnetic stir bar **116** may take various other shapes (e.g., rectangle, X-shape, triangle, helix, donut, etc.) without departing from the scope hereof. In some embodiments, magnetic stir bar **116** may extend vertically along the length of mixing region **104a** (see FIG. 3), which may aid in uniform agitation of substance **106a** and substance **106b**. For example, the magnetic stir bar **116** may have a length that is about 25% of the length of mixing region **104a**, about 50% of the length of mixing region **104a**, about 75% of the length of mixing region **104a**, or about 100% the length of mixing region **104a**. Such an instance of magnetic stir bar **116** may have any of the aforementioned shapes discussed above. In some embodiments, magnetic stir bar **116** may comprise a polytetrafluoroethylene (PTFE) coated magnet (e.g., PTFE coated alnico). Other materials are within the scope hereof including, but not limited to, samarium cobalt, neodymium, or ferrite. Glass coatings may also be employed. Generally, magnetic stir bar **116** may comprise any magnetic material such that magnetic stir bar **116** is rotatable by the rotating magnetic field **114**. Although magnetic stir bar **116** is referred to herein as “magnetic,” embodiments are contemplated where magnetic stir bar **116** is ferromagnetic rather than a permanent magnet or conductive such that magnetic properties are created by induced magnetism (for example, electromagnetism or induced currents).

(22) In some embodiments, the location of magnetic drive **112** along the length of mixing region **104a** is adjustable (the length being measured from top surface **104b** to bottom surface **104c**). In some embodiments, the position of magnetic drive **112** is manually adjustable, and the operator may adjust the magnetic drive **112** vertically along the length of the mixing vessel **102**, such that the magnetic drive **112** may surround the mixing region **104a** at different vertical locations. As shown, magnetic drive **112** may have an inner surface that is in contact with an exterior surface of mixing vessel **102** and may be held in place by friction and/or by other coupling means, such as fasteners, adhesives, or the like. In such instances, the magnetic drive **112** will generate a rotating magnetic field **114** as disclosed and control agitation via the magnetic stir bar **116** along the common plane at the vertical position of the magnetic drive **112**. Thus, vertical adjustment of the magnetic drive **112** allows improved control of mixing substance **106a** and substance **106b**, especially of mixing volumes that occupy at least 50% of the mixing region **104a**. It is contemplated that the magnetic drive **112** may itself be actuated by a drive system or the like to adjust the position along mixing vessel **102** during mixing, for example, powered by a motor to slide along the length of mixing vessel **102**.

(23) The magnetic drive **112** may be controlled by a computer or controller **118a**, thereby enabling control of the rotating magnetic field **114**. In some embodiments, controller **118a** is configured to

control the speed and/or the direction of the rotating magnetic field **114**. Controller **118a** may comprise a processor **118b** and a memory **118c**. Memory **118c** may be configured to store non-transitory computer-readable media executable by processor **118b**. For example, memory **118c** may store executable software instructions for controlling the speed and/or direction of the rotating magnetic field **114**. Generally, memory **118c** may store computer-executable instructions executable by processor **118b** to carry out any of the functionality of the systems described here. For example, controller **118a** may further control pump **110**.

(24) Additionally, in some embodiments, controller **118a** may include or otherwise control at least one H-bridge circuit **120**. In some such embodiments, the at least one H-bridge circuit **120** is configured to control the direction (for example, by reversing the polarity of an electromagnetic circuit) and/or the speed of rotation of rotating magnetic field **114**. Control of the speed may be accomplished by varying the voltage applied across a load via the at least one H-bridge circuit **120** as will be appreciated by one of skill in the art. To alter the direction in which voltage is supplied, the at least one H-bridge circuit **120** may control the series of switches that deliver power to the load. Hall effect sensors may also be employed to monitor the position of magnetic stir bar **116** as it rotates so as to appropriately modulate the energization of the coils of magnetic drive **112** to produce rotation of magnetic stir bar **116** at a speed controllable by controller **118a**.

(25) Mixing system **100** may further include a heating element **122** to control the temperature of the mixing vessel **102**. Specifically, heating element **122** may heat substance **106a** and substance **106b** to a temperature necessary for process reactions to proceed and may further aid in the efficiency of chemical reactions, filtrations, and crystallizations. Although heating element **122** is described herein, it should be understood that a cooling jacket could be employed in a similar fashion to reduce the temperature of mixing vessel **102** if desired. For example, a cooling jacket comprising a plurality of fluid pathways for coolant to flow around mixing vessel **102** could be employed to control the temperature of exothermic mixing reactions. As previously discussed, one advantage of the embodiments described herein is the ability to pressurize mixing vessel **102** such that the allowable operating temperatures may be higher than when operating at atmospheric pressure. The heating element **122** may surround the mixing vessel **102** for temperature control in some embodiments. In such embodiments, the heating element **122** may be an electric heat source, such as a heater jacket coupled to an exterior of the mixing vessel **102**. The heating element **122** may be between the magnetic drive **112** and the mixing vessel **102** such that the magnetic drive **112** may surround the heating element **122**. Heating element **122** may extend along any portion of the length of mixing vessel **102**. In some embodiments, heating element **122** extends along 25% of the length of mixing vessel **102** or mixing region **104a**, along 50% of the length of mixing vessel **102** or mixing region **104a**, along 75% of the length of mixing vessel **102** or mixing region **104a**, or along 100% of the length of mixing vessel **102** or mixing region **104a**.

(26) As can be seen in FIG. 1C, the heating element **122** may be concentric with the mixing vessel **102**. Furthermore, the heating element **122** may also be concentric with the magnetic drive **112**, with the magnetic drive **112** surrounding the heating element **122**. Thus, magnetic drive **112** may also be concentric with the mixing vessel **102**.

(27) As previously discussed, in some embodiments, the mixing vessel **102** is a filter reactor with a filtration system. The filtration system may include a filter **124a** and at least one bottom outlet valve **124b**. As some chemical reactions result in the generation of byproducts, such as solid byproducts occurring via crystallization or precipitation within the solution, the filter **124a** may retain these byproducts without impeding the ongoing continuous process. Additionally, it may be desirable to filter out the byproducts for later collection. Advantageously, the filter **124a** also prevents the need for filtration in a separate piece of laboratory equipment. Moreover, as no transfer of solution occurs outside of the filter reactor, there is little to no transfer loss associated with a mixing vessel **102** configured to conduct both chemical reactions in the mixing region **104a** and filtration via the filtration system. Thus, a secondary, separate filter system is not needed. As

such, filter reactors may allow for more efficient chemical reactions and increased yields of end products. Embodiments featuring a filtration system may permit improvements in process efficiency and eliminate the need for operator manual interference as will be appreciated by one of skill in the art. The filtration system may be included within mixing vessel **102** below magnetic stir bar **116** such that it does not interfere with the filtering. For example, a step may be included in the interior shell of mixing vessel **102** to retain magnetic stir bar **116** above the filter media and allow accumulation of filtrate at the bottom of mixing vessel **102**.

(28) Mixing system **100** may further comprise one or more sensors, such as sensors **126**, which may be coupled to the mixing vessel **102** to monitor the operating conditions thereof. These sensors **126** may monitor various conditions, including but not limited to temperature, pressure, pH, volume, density, color, and opacity. In such embodiments, sensors may be utilized to detect the operating conditions of mixing vessel **102** throughout the duration of the reaction within mixing system **100**. Monitoring of such conditions may occur during pressurization of mixing vessel **102**, during the chemical reaction of substance **106a** and substance **106b**, during the filtration process of a resulting byproduct, or at any other point in time during the operation of mixing system **100**. Sensors **126** may be communicatively coupled to controller **118a** to transfer the sensor data thereto.

(29) With specific reference to FIG. **1C**, magnetic drive **112** is described in further detail in accordance with embodiments of the present disclosure. As shown, the magnetic drive **112** may include a stator **128**, where the stator may be made up of a plurality of instances of stator teeth **130** and a plurality of instances of stator windings **132** wrapped around the stator teeth **130**. The stator windings **132** may be selectively energized to establish an electromagnetic field, i.e., the rotating magnetic field **114**. The timing and voltage of the energization of stator windings **132** control the strength and rotation of the rotating magnetic field **114**.

(30) The configuration of stator windings **132** around the stator teeth **130** may control the capacity and activity of the magnetic drive **112** where winding configuration is defined by how many stator teeth are present, which stator teeth **130** the stator windings **132** are located on as well as the how many loops of conductor make up each of stator windings **132**. In some embodiments, the stator windings **132** may wrap around the stator teeth in at least 200 loops. Other embodiments may feature stator windings **132** that wrap around the stator teeth in less than 200 loops. Some other embodiments may utilize stator windings that wrap around the stator teeth in more than 200 loops.

(31) In some embodiments, stator **128** includes six instances of stator teeth **130** on the stator **128**. Other embodiments may have a stator design featuring more or less than six instances of stator teeth **130** on the stator **128**. In some embodiments, the stator **128** includes a number of instances of stator teeth **130** that is a multiple of six, such as six, twelve, eighteen, or twenty-four instances of stator teeth **130**. In some embodiments, stator **128** has a concentrated winding stator configuration that features a stator **128** with winding loops wrapped around at least one individual tooth of stator teeth **130**. In some embodiments, stator windings **132** are wound only around alternating stator teeth **130**. Other embodiments may utilize a stator **128** configuration where stator windings **132** are wrapped around every stator tooth of the stator **128**. Still some embodiments may be employed such that a combination of alternating and sequential teeth of stator teeth **130** are wrapped with the stator windings **132** without departing from the specification hereof. For instance, as depicted, two consecutive stator teeth may have stator windings **132**, followed by a third stator tooth that does not have any stator windings **132**. This winding pattern may then continue for the remaining stator teeth of the stator **128**.

(32) As previously discussed, stator **128** may be coupled to the exterior of mixing vessel **102**. For example, an inner surface **134** of stator **128** may directly contact the exterior surface of mixing vessel **102**. When a heating element **122** is used, the inner surface **134** of stator **128** may be in contact with the heating element **122**. Placing the stator **128** (and, thereby, the rotating magnetic field **114**) as close to the magnetic stir bar **116** as possible may be advantageous, as the torque supplied may increase with decreasing distance between stator **128** and magnetic stir bar **116**. The

stator **128** may be made from any suitable material, including conventional or electrical steel, iron, steel, or fiberglass. In some embodiments, the stator **128** is formed via an additive manufacturing process.

(33) Turning now to FIG. 2., a second example of a mixing system **200** is illustrated in accordance with embodiments of the present disclosure. Mixing system **200** may be substantially similar to mixing system **100** discussed above. Mixing system **200** may include a mixing vessel **202**, a mixing region **204**, inlet ports **206**, and a heating element **208**. Mixing system **200** may include any of the components described above with respect to FIGS. 1A-1C.

(34) In contrast to mixing system **100**, mixing system **200** comprises first brushless magnetic drive **210a** and second brushless magnetic drive **210b**, both surrounding the mixing vessel **202**, such that the first brushless magnetic drive **210a** is located above the second brushless magnetic drive **210b**. As shown in FIG. 2, first brushless magnetic drive **210a** and second brushless magnetic drive **210b** may be stacked directly on top of each other such that the top surface of second brushless magnetic drive **210b** contacts the bottom surface of first brushless magnetic drive **210a**. In some embodiments, first brushless magnetic drive **210a** and second brushless magnetic drive **210b** are spaced apart along the length of mixing vessel **202**. The presence of more than one brushless magnetic drive may provide the advantage of solution mixing along a greater length of the mixing vessel **202**. This may be especially useful for embodiments with a mixing region **204** that receives mixing substances at or near the maximum fill capacity of the mixing vessel **202**.

(35) First brushless magnetic drive **210a** and second brushless magnetic drive **210b** may stir in tandem, e.g., operating at the same speed and/or in the same direction. In some embodiments, first brushless magnetic drive **210a** and second brushless magnetic drive **210b** operate at different speeds and/or rotate in different directions. Each of first brushless magnetic drive **210a** and second brushless magnetic drive **210b** may generate a distinct magnetic field. First brushless magnetic drive **210a** and second brushless magnetic drive **210b** may be controlled by a control system, such as controller **118a** described above.

(36) In some embodiments, each of first brushless magnetic drive **210a** and second brushless magnetic drive **210b** is configured to rotate a separate stir bar. For example, and as shown, first brushless magnetic drive **210a** rotates a first magnetic stir bar **212a**, and second brushless magnetic drive **210b** rotates a second magnetic stir bar **212b**. Mixing vessel **202** may, in some embodiments, include an internal support configured to support or hold the first magnetic stir bar **212a** within the mixing region **204** at a vertical location equivalent or above a vertical location of a bottom surface of the first brushless magnetic drive **210a** bottom surface. Thus, when first brushless magnetic drive **210a** is not in operation, first magnetic stir bar **212a** will not fall to the bottom of mixing region **204** by being supported by the internal support, and when first brushless magnetic drive **210a** is activated, first magnetic stir bar **212a** may be in the horizontal plane of first magnetic stir bar **212a** and the magnetic field generated by first magnetic stir bar **212a** such that first magnetic stir bar **212a** can be rotated as described above. The internal support may project from an internal wall of mixing vessel **202** or may extend from a bottom or top of mixing region **204**. For example, it is contemplated that first magnetic stir bar **212a** and second magnetic stir bar **212b** may be mounted to a vertical shaft (see, e.g., FIG. 3) within mixing region **204** and mounted at a location along the shaft that is in-line with the corresponding drive (e.g., first brushless magnetic drive **210a** or second brushless magnetic drive **210b**). In some embodiments, more than one drive is employed to drive a single stir bar, such as when a stir bar has a length that is greater than the length of a single brushless magnetic drive (see, e.g., FIG. 3). Advantageously, in some embodiments, the magnetic fields of first brushless magnetic drive **210a** and second brushless magnetic drive **210b** may be rotated in different directions. Spinning multiple magnetic stir bars in opposite directions may increase the agitation efficiency of the mixing system **200**.

(37) FIG. 3 depicts a third example of a mixing system **300** for some embodiments. Mixing system **300** may be substantially similar to the mixing system **100** and mixing system **200** discussed above

and may include a double-walled mixing vessel **302**, a mixing region **304**, inlet ports **306**, and a brushless magnetic drive **308** that generates a rotating magnetic field (not shown) to drive a magnetic stir bar **310**. In some embodiments, the magnetic stir bar **310** may be, as shown in FIG. 3, a vertical magnetic stir bar that spans a vertical length within the mixing region **304** (e.g., a portion of the overall length of mixing region **304** or the entire length). In other embodiments, the stir bar may be more similar to the horizontally oriented stir bars depicted in FIGS. 1B and 2.

(38) In contrast to mixing system **100** and mixing system **200**, mixing system **300** is configured as a double-walled mixing vessel **302**, where a first wall **312a** occupies the same diameter as the mixing region **304**, i.e., defines the mixing volume. A second wall **312b** may be concentric with both the mixing region **304** and the first wall **312a** of the double-walled mixing vessel **302**. The second wall **312b** may have a minimum diameter that is at least larger than the diameter of the first wall **312a** such that a gap is defined between the first wall **312a** and the second wall **312b**. At least one instance of brushless magnetic drive **308** may be coupled to the outside of the (outer) second wall **312b**, as shown. At least one instance of brushless magnetic drive **308** may generate a magnetic field to rotate a magnetic stir bar **310** within double-walled mixing vessel **302**, as previously discussed.

(39) The double-walled mixing vessel **302** may employ a heating element **314** that provides heating (or cooling) between the first wall **312a** and second wall **312b**. Specifically, heating element **314** may be a recirculating heater (or chiller) that recirculates fluid between the first wall **312a** and the second wall **312b** of the double-walled mixing vessel **302** for either heating or cooling of the double-walled mixing vessel **302**. Notably, the double-walled mixing vessel **302** and, thus, the heating element **314** may extend the same length as the mixing region **304**. However, in some other embodiments, the length of the heating element **314** may only extend over a length less than the overall length of the double-walled mixing vessel **302**. In such instances, the heating element **314** may be designed to align with the length of the mixing region **304** occupied by some pre-established volume of mixing substances.

(40) The heating element **314**, which may be a recirculating heater that recirculates fluid between a first wall **312a** and a second wall **312b** of a double-walled mixing vessel **302**, may also provide mixing system **300** with an external layer to pull a vacuum on for filtration purposes. The vacuum may provide insulation of the double-walled mixing vessel **302** and the one or more mixing substances being mixed or filtered therewithin. Insulation can assist in reducing the load on temperature control and in further increasing the efficiency of the chemical process.

(41) It will be appreciated that, while not depicted in FIG. 3, multiple instances of brushless magnetic drive **308** surrounding the second wall **312b** of the double-walled mixing vessel **302** may be stacked directly on top of each other, such as shown in FIG. 2. Still other embodiments may feature multiple instances of brushless magnetic drive **308** surrounding the second wall **312b** of the double-walled mixing vessel **302** but with some vertical distance between the instances of brushless magnetic drive **308**. As is the case in FIG. 1B, a single instance of brushless magnetic drive **308** may be adjusted by an operator (or automatically) vertically along the length of the double-walled mixing vessel **302** such that the brushless magnetic drive **308** may surround the mixing region at a plurality of vertical locations.

(42) Features described above as well as those claimed below may be combined in various ways without departing from the scope hereof. The following examples illustrate some possible, non-limiting combinations: Clause 1. A mixing system, comprising: a mixing vessel, comprising: a fluid inlet; a mixing region configured to receive a first mixing substance and a second mixing substance via the fluid inlet; and at least one outlet fluidly connected to the mixing region; a magnetic stir bar disposed within the mixing vessel; a brushless magnetic drive that surrounds the mixing vessel and that is configured to generate a rotating magnetic field, thereby rotating the magnetic stir bar to mix the first mixing substance with the second mixing substance; and a controller configured to control a speed and a direction of the rotating magnetic field. Clause 2. The mixing system of clause 1,

further comprising a heating element configured to heat the mixing vessel, wherein the heating element surrounds the mixing vessel and extends along an exterior length of the mixing vessel. Clause 3. The mixing system of clause 1 or clause 2, wherein the mixing region has a volume less than 100 milliliters. Clause 4. The mixing system of any of clauses 1-3, further comprising a pump coupled to the mixing vessel and operable to pressurize the mixing vessel. Clause 5. The mixing system of any of clauses 1-4, wherein the brushless magnetic drive is adjustable vertically along a length of the mixing vessel to adjust a vertical position of the magnetic stir bar. Clause 6. The mixing system of any of clauses 1-5, wherein the brushless magnetic drive is a first brushless magnetic drive, the rotating magnetic field is a first rotating magnetic field, and the mixing system further comprises: a second brushless magnetic drive surrounding the mixing vessel and configured to generate a second rotating magnetic field. Clause 7. The mixing system of any of clauses 1-6, wherein the magnetic stir bar is a first magnetic stir bar and the mixing system further comprises: a second magnetic stir bar that is rotated by the second rotating magnetic field. Clause 8. A system for vessel mixing, comprising: a pressurizable mixing vessel, comprising: at least one inlet; a mixing region configured to receive a first mixing substance and a second mixing substance via the at least one inlet; and at least one outlet fluidly connected to the mixing region; a magnetic stir bar disposed within the pressurizable mixing vessel; and a brushless magnetic drive surrounding the pressurizable mixing vessel and configured to generate a rotating magnetic field, thereby rotating the magnetic stir bar to mix the first mixing substance with the second mixing substance. Clause 9. The system of clause 8, further comprising a pump configured to pressurize the pressurizable mixing vessel. Clause 10. The system of clause 8 or clause 9, further comprising: a heating element configured to heat the pressurizable mixing vessel, wherein the heating element surrounds the pressurizable mixing vessel and extends along a length of the pressurizable mixing vessel. Clause 11. The system of any of clauses 8-10, wherein the magnetic stir bar is vertically oriented within the pressurizable mixing vessel. Clause 12. The system of any of clauses 8-11, wherein the brushless magnetic drive further comprises: a stator comprising a plurality of stator teeth; and a plurality of windings wound around the plurality of stator teeth, wherein selective energization of the plurality of windings around the plurality of stator teeth generates the rotating magnetic field that causes rotation of the magnetic stir bar. Clause 13. The system of any of clauses 8-12, wherein the pressurizable mixing vessel is a filter reactor, further comprising: a filter configured to filter a solid material resulting from a reaction of the first mixing substance with the second mixing substance in the mixing region, wherein a volume of the mixing region is in a range of 1 milliliter to 20 milliliters. Clause 14. The system of any of clauses 8-13, wherein the brushless magnetic drive is adjustable vertically along a length of the pressurizable mixing vessel to adjust a vertical position of the magnetic stir bar. Clause 15. A system for planar vessel mixing, comprising: a mixing vessel, comprising: a fluid inlet; a mixing region configured to receive a first mixing substance and a second mixing substance via the fluid inlet; and at least one outlet fluidly connected to the mixing region; a magnetic stir bar disposed within the mixing vessel; a first brushless magnetic drive surrounding the mixing vessel and operable to generate a first magnetic field that rotates a first magnetic stir bar to thereby mix the first mixing substance and the second mixing substance; and a second brushless magnetic drive that surrounds the mixing vessel. Clause 16. The system of clause 15, further comprising: a second magnetic stir bar, wherein the second brushless magnetic drive is configured to generate a second magnetic field that rotates the second magnetic stir bar. Clause 17. The system of clause 15 or clause 16, further comprising: a controller comprising at least one H-bridge and operable to rotate the first magnetic field and the second magnetic field to control a rotational speed and direction of the first magnetic stir bar and of the second magnetic stir bar. Clause 18. The system of any of clauses 15-17, further comprising: a heating element configured to heat the mixing vessel, wherein the heating element surrounds the mixing vessel and extends along a length of the mixing vessel. Clause 19. The system of any of clauses 15-18, wherein the heating element is an electric heating element coupled to an exterior of

the mixing vessel, wherein at least one of the first brushless magnetic drive or the second brushless magnetic drive surrounds the electric heating element. Clause 20. The system of any of clauses 15-19, wherein the mixing vessel is a double-walled mixing vessel, wherein the heating element is a recirculating heater that recirculates fluid between a first wall and a second wall of the double-walled mixing vessel.

(43) Although the present disclosure has been described with reference to the embodiments illustrated in the attached drawing figures, it is noted that equivalents may be employed, and substitutions made herein without departing from the scope of the present disclosure as recited in the claims.

Claims

1. A mixing system, comprising: a mixing vessel, comprising: a fluid inlet; a mixing region configured to receive a first mixing substance and a second mixing substance via the fluid inlet; and at least one outlet fluidly connected to the mixing region; a first magnetic stir bar disposed within the mixing vessel; a first brushless magnetic drive that surrounds the mixing vessel and that is configured to generate a first rotating magnetic field, thereby rotating the first magnetic stir bar; a second magnetic stir bar disposed within the mixing vessel; a second brushless magnetic drive that surrounds the mixing vessel and that is configured to generate a second rotating magnetic field that rotates the second magnetic stir bar, wherein rotating at least one of the first magnetic stir bar and the second magnetic stir bar mixes the first mixing substance with the second mixing substance; and a controller configured to control a speed and a direction of at least one of the first rotating magnetic field or the second rotating magnetic field.
2. The mixing system of claim 1, further comprising a heating element configured to heat the mixing vessel, wherein the heating element surrounds the mixing vessel and extends along an exterior length of the mixing vessel.
3. The mixing system of claim 1, wherein the mixing region has a volume less than 100 milliliters.
4. The mixing system of claim 1, further comprising a pump coupled to the mixing vessel and operable to pressurize the mixing vessel.
5. The mixing system of claim 1, wherein the first brushless magnetic drive is adjustable vertically along a length of the mixing vessel to adjust a vertical position of the first magnetic stir bar.
6. The mixing system of claim 1, wherein the second brushless magnetic drive is adjustable vertically along a length of the mixing vessel to adjust a vertical position of the second magnetic stir bar.
7. A system for vessel mixing, comprising: a pressurizable mixing vessel, comprising: at least one inlet; a mixing region configured to receive a first mixing substance and a second mixing substance via the at least one inlet; and at least one outlet fluidly connected to the mixing region; a first magnetic stir bar disposed within the pressurizable mixing vessel; a first brushless magnetic drive surrounding the pressurizable mixing vessel and configured to generate a first rotating magnetic field, thereby rotating the first magnetic stir bar; a second magnetic stir bar disposed within the pressurizable mixing vessel; and a second brushless magnetic drive surrounding the pressurizable mixing vessel and configured to generate a second rotating magnetic field, thereby rotating the second magnetic stir bar, wherein rotating at least one of the first magnetic stir bar and the second magnetic stir bar mixes the first mixing substance with the second mixing substance.
8. The system of claim 7, further comprising a pump configured to pressurize the pressurizable mixing vessel.
9. The system of claim 7, further comprising: a heating element configured to heat the pressurizable mixing vessel, wherein the heating element surrounds the pressurizable mixing vessel and extends along a length of the pressurizable mixing vessel.
10. The system of claim 7, wherein the first magnetic stir bar is vertically oriented within the

pressurizable mixing vessel.

11. The system of claim 7, wherein the first brushless magnetic drive further comprises: a stator comprising a plurality of stator teeth; and a plurality of windings wound around the plurality of stator teeth, wherein selective energization of the plurality of windings around the plurality of stator teeth of the first brushless magnetic drive generates the first rotating magnetic field that causes rotation of the first magnetic stir bar.

12. The system of claim 7, wherein the pressurizable mixing vessel is a filter reactor comprising: a filter configured to filter a solid material resulting from a reaction of the first mixing substance with the second mixing substance in the mixing region, wherein a volume of the mixing region is in a range of 1 milliliter to 20 milliliters.

13. The system of claim 7, wherein the first brushless magnetic drive is adjustable vertically along a length of the pressurizable mixing vessel to adjust a vertical position of the first magnetic stir bar.

14. The system of claim 7, wherein the second brushless magnetic drive is adjustable vertically along a length of the pressurizable mixing vessel to adjust a vertical position of the second magnetic stir bar.

15. A system for planar vessel mixing, comprising: a mixing vessel, comprising: a fluid inlet; a mixing region configured to receive a first mixing substance and a second mixing substance via the fluid inlet; and at least one outlet fluidly connected to the mixing region; a first magnetic stir bar disposed within the mixing vessel; a first brushless magnetic drive surrounding the mixing vessel and operable to generate a first magnetic field that rotates the first magnetic stir bar; a second magnetic stir bar disposed within the mixing vessel; and a second brushless magnetic drive surrounding the mixing vessel and operable to generate a second magnetic field that rotates the second magnetic stir bar, wherein rotating at least one of the first magnetic stir bar or the second magnetic stir bar mixes the first mixing substance with the second mixing substance.

16. The system of claim 15, further comprising: a controller comprising at least one H-bridge and operable to rotate the first magnetic field and the second magnetic field to control a rotational speed and a direction of the first magnetic stir bar and of the second magnetic stir bar.

17. The system of claim 15, further comprising: a heating element configured to heat the mixing vessel, wherein the heating element surrounds the mixing vessel and extends along a length of the mixing vessel.

18. The system of claim 17, wherein the heating element is an electric heating element coupled to an exterior of the mixing vessel, wherein at least one of the first brushless magnetic drive or the second brushless magnetic drive surrounds the electric heating element.

19. The system of claim 17, wherein the mixing vessel is a double-walled mixing vessel, wherein the heating element is a recirculating heater that recirculates fluid between a first wall and a second wall of the double-walled mixing vessel.

20. The system of claim 15, further comprising: a sensor configured to monitor a position of the first magnetic stir bar and a position of the second magnetic stir bar.
